Homework 10 - IIR filters

Deadline: 4th test date (4th of June)
The mandatory questions account for 100 points, while the optional questions together contribute up to an additional 50 points.

1. Outline the different classes of filters in terms of the frequency regions where they attenuate the amplitude of the oscillations. Draw the corresponding frequency response curves.
2. What is the passband of a filter? How is it different from the stopband?
3. What do we mean by the cut-off frequency of a filter?
4. Explain the concept of rolloff in filter design. How is it measured, and what does it indicate about a filter's performance?
5. Define the following terms in the context of filter design: passband, stopband, cutoff frequency, bandwidth, ripple, rolloff, insertion loss, and stop-band attenuation.
6. Explain the significance of the order of a filter. How does it relate to the number of poles in the z-domain?
7. What is ripple in the context of filter design? How does it affect the filter's performance in the passband and stopband?
8. Describe the trade-offs involved in filter design between the transition band, filter order, and ripple amplitude.
9. Explain the concept of a chirp signal. Sketch its waveform and also the effect of different types of filters on it (low-, high-, bandpass).
10. What are the different approaches to designing an IIR filter?
11. Describe the process of digitizing an analog filter. What are the common methods used for this purpose?
12. What is the significance of the Nyquist frequency in digital filter design? How does it relate to the sampling rate? How is it represented in the z-plane?
13. Design a lowpass filter for signals sampled at 16 Hz such that the direct current gain is unity and the cutoff frequency is at 4Hz.

    Hint: Use the direct design method with a single zero and pole. Check the course material.
14. What is a notch filter? What parameters are critical in defining the filter's characteristics?
15. Explain the concept of notch depth in the context of notch filters. How is it related to stop-band attenuation?
16. Design a notch filter with notch frequency of 20 Hz for signals sampled at 160 Hz. Attenuation should be of at least 40 dB and the bandwidth of 4 Hz. The gain in the passband should be unity.

    Hint: Use the direct design method with two poles and two zeros. Check the course material.
17. Implement a Python script to compute and plot the frequency response of the IIR filter designed in the previous question.
18. Explain the bilinear transform method for designing digital filters from analog prototypes. Provide an example.
19. Derive the bilinear transformation formula.
20. Design a low-pass IIR filter with a cutoff frequency of 5 Hz and a sampling frequency of 20 Hz using the bilinear transform method. Provide the filter coefficients.

    Hint: Use the direct design method with a single zero and pole. Check the course material.
21. Describe the concept of pre-warping in the bilinear transform method. Why is it necessary and how is it applied?
22. Estimate the correction in the cutoff frequency ω_c for a sampling rate of 100 Hz using the pre-warping method.
23. Derive up to second order the pre-warp correction using the MacLaurin series.
24. Explain the impulse-invariant method for designing IIR filters. How does it work and what are its main advantages and disadvantages?
25. Given the transfer function $H(z) = \frac{0.465z}{z-0.535}$, verify the correctness of this representation for a low-pass filter designed using the impulse-invariant method.
26. Describe the process of mapping poles and zeros from the s-plane to the z-plane. Why is this important in digital filter design?